5.2 #s: 2, 3, 6, 13, 14

5.3 #s: 3, 5, 9 (use exercise 5.2.2)

5.4 #s: 9

by Tyler Trotter

Math 3210 Homework 9

2: If f is a bounded function defined on a closed, bounded interval $[a,b]$ and if f is integrable on each interval $[a,r]$ with $a<r<b$, then prove if f is integrable on $[a,b]$ and

$$\int_a^b f(x)\,dx = \lim_{r\to b}\int_a^r f(x)\,dx$$

Observe that the analogous result holds if $[a,r]$ is replaced by $[r,b]$ in the hypothesis and in the integral on the right and the limit taken as $r\to a$. Hint: Use theorem 5.2.3 and exercise 5.1.8

Since we are holding f to be integrable on each closed, bounded interval $[a,r]$ and defined over the closed, bounded interval $[a,b]$. Since $a<r<b$, it clearly follows from Additivity that r may draw arbitrarily close to b and since $f(b)$ is defined, $f(x)$ is defined and integrable for any interval $\lim_{r\to b}[a,r]$. Thus, it suffices to say

$$\int_a^b f(x)\,dx = \lim_{r\to b}\int_a^r f(x)\,dx.$$

$\lim_{r\to b} r = b$, even though $r<b$.

Likewise $m(r-a) \le \underline{\int_a^r} f(x)\,dx \le \overline{\int_a^r} f(x)\,dx \le M(r-a)$ implies the integral of $f(x)$ is bounded over the domain by some lower and upper rectangle of width $r-a$. With $\lim_{r\to b}$, we get

$$\lim_{r\to b} m(r-a) \le \lim_{r\to b}\underline{\int_a^r} f(x)\,dx \le \lim_{r\to b}\int_a^r f(x)\,dx \le \lim_{r\to b} M(r-a).$$

becomes

$$m(b-a) \le \underline{\int_a^b} f(x)\,dx \le \overline{\int_a^b} f(x)\,dx \le M(b-a)$$ meaning the integral exists. ∎

5.2 #s: 3, 6, 13, 14
5.3 #s: 3, 5, 9 (use Ex. 5.2.2)
5.4 #s: 9

Homework 7, cont

3: Prove Theorem 5.2.10. That is, prove that if f is a bounded function on a bounded interval $[a,b]$ and f is continuous except at finitely many points in $[a,b]$, then f is integrable on $[a,b]$. Hint: Use the preceding exercise, interval additivity, and an induction argument on the number of discontinuities.

By the previous exercise, we learn $\int_a^b f(x)dx = \lim_{r\to b} \int_a^r f(x)dx$. This means for any b, $f(b)$ does not matter to the value of the integral. Thus, for any finite number of discontinuities of f (aka b's) then we may consider any such interval $[a,r]$ over which f is continuous and let c_n be any discontinuity. Then

$$\lim_{r_1\to c_1}\int_a^{r_1} f(x)dx + \lim_{r_2\to c_2}\int_{c_1}^{r_2} f(x)dx + \dots + \lim_{r_n\to c_n}\int_{c_{n-1}}^{r_n} f(x)dx + \int_{c_n}^{b} f(x)dx$$

Base Case: $n=1$, then we've proven that in the previous problem

Inductive Step: Assume $\int_a^b f(x)dx = \lim_{r_1\to c_1}\int_a^{r_1} f(x)dx + \lim_{r_2\to c_2}\int_{c_1}^{r_2} f(x)dx \dots$

Need to show: $\int_a^b f(x)dx$ still exists by adding a discontinuity.

$$\lim_{r_1\to c_1}\int_a^{c_1} f(x)dx + \lim_{r_2\to c_2}\int_{c_1}^{r_2} f(x)dx + \lim_{r_n\to c_n}\int_{c_{n-1}}^{r_n} f(x)dx \quad \lim_{r_{n+1}\to c_{n+1}}\int_{c_n}^{r_{n+1}} f(x)dx + \int_{c_n}^{b} f(x)dx$$

We have assume the function's integral has held for n discon.

Now, we claim what remains: $\int_{c_n}^{r_{n+1}} f(x)dx$ is just another instantiation of the base case.

So, $\int_a^b f(x)dx$ exists so long as only finitely many discontinuities exist. Likewise, the other half of the integral $\lim_{r\to d_n}\int_{r_n}^{d_{n+1}} f(x)dx$ is similar logic as above.

5.2 #s: 6, 13, 14
5.3 #s: 3, 5, 9 (use 3.2.2)
5.4 #s: 7

Homework 9, cont.

6: Prove that $1 \leq \int_{-1}^{1} \frac{1}{1+x^{2n}}\, dx \leq 2 \quad \forall n \in \mathbb{N}$.

$$\int_{-1}^{1} \frac{1}{1+x^{2n}}\, dx = \ln(1+x^{2n})$$

We know from Ex. 5.1.8, $m(1-(-1)) \leq \int_{-1}^{1} \frac{1}{1+x^{2n}}\, dx \leq \int_{-1}^{1} \frac{1}{1+x^{2n}}\, dx \leq M(1-(-1))$
Which is $2m \leq \ldots \leq 2M$.
To find m and M, we consider $\inf \left\{ \frac{1}{1+x^{2n}} : x \in [-1,1] \right\}$.
Since x is raised to the power of $2n$, x^{2n} will always be positive and $x \in [-1,0) = x \in (0,1]$ for any $|x|$ as a result.
So, the minimum of $f(x)$ on $[-1,1]$ will be where $1+x^{2n}$ is maximized. Adding anything more than 0 will make this larger, so we choose the largest element of $[-1,1]$. That is either $(1)^{2n}$ or $(-1)^{2n}$. However, for any n, this will be 1. So, $1+x^{2n} \leq 2$. And $\frac{1}{2} \leq \frac{1}{1+x^{2n}}$. Thus, $2(\frac{1}{2}) \leq \int \frac{1}{1+x^{2n}} dx$.
Now for M, we find $\sup \left\{ \frac{1}{1+x^{2n}} : x \in [-1,1] \right\}$. The sup occurs when $1+x^{2n}$ is minimized. This occurs when x^{2n} is minimized. For $x \in [-1,1]$, this occurs at $x=0$. So, $1 \leq 1+x^{2n}$ and $\frac{1}{1+x^{2n}} \leq 1$ and thus $\int_a^b \frac{1}{1+x^{2n}}\, dx \leq 2(1)$.

To conclude, we have found the following inequality:

$$1 \leq \int_{-1}^{1} \frac{1}{1+x^{2n}}\, dx \leq 2 \quad \text{for any } n \in \mathbb{N}.$$

Which is what we wanted to prove. ∎

5.2 #s: 13, 14
5.3 #s: 3, 5, 9 (use Ex 5.2.2)
5.4 #s: 9

13: Let $\{f_n\}$ be a sequence of integrable functions defined on a closed, bounded interval $[a,b]$. If $\{f_n\}$ converges uniformly on $[a,b]$ to a function f, prove that f is integrable and
$$\int_a^b f(x)\,dx = \lim \int_a^b f_n(x)\,dx.$$

Intuitively, this means the lim area under the curve for all functions f_n is the area under the curve of f.

By definition $\{f_n\}$ converges uniformly if for all $\varepsilon>0$, $\exists N>0$ s.t. $\forall n>N$ then $|f_n - f| < \varepsilon$ for all $x \in D$.

Now consider two Theorems from 5.2: 5.2.3 and 5.2.6 ... for any N, $n>N$ implies
It thus follows that
that $|f_n - f| <$

5.2 #s: 14
5.3 #s: 3, 5, 9 (use Ex 5.2.2)
5.4 #s: 9

14: Is the function which is $\sin(\frac{1}{x})$ for $x \neq 0$ and 0 for $x=0$ integrable on $[0,1]$? Justify your answer.

$$f(x) = \begin{cases} \sin(\frac{1}{x}) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Suppose f is discontinuous at $x=0$ (as $\sin$ is cont. everywhere else). By Theorem 5.2.10, f is a bounded function on $[0,1]$ which is a closed, bounded interval, and f is continuous everywhere except maybe at $x=0$, then f is integrable on $[0,1]$. ∎

5.3 # 3: Find $\frac{d}{dx}\int_0^{2x} \sin t^2 \, dt$

$F(x) = \int_0^{2x} \sin t^2 \, dt$; $\frac{dF(x)}{dx} = \frac{d}{dx}\int_0^{2x} \sin(t^2)\, dt$

By Fundamental Theorem of Calculus, $\frac{d}{dx}\int_0^{2x} \sin(t^2)\,dt$ is $\sin[(2x)^2] = \sin(4x^2)$.

5: